SCALE – additivity and burst-factor analysis (preview)

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1 Evaluation (draft)

1.1 Additivity of Capacity Units (RQ3)

SCALE charges every request r its Capacity Units $SK_r = SK_{max}/MST_r$ and admits a workload as long as the Capacity Units consumed per minute do not exceed SK_{max}. For a mix in which class i represents a share p_i of the requests, this accounting predicts a capacity

$$C_{\text{mix}} = \Big(\sum_i \frac{p_i}{C_i}\Big)^{-1}, \tag{1}$$

i.e., the weighted *harmonic* mean of the class capacities C_i. This is the utilisation law of operational analysis, in which $1/C_i$ is the service demand of class i [?]. The law is classical; what is not obvious is whether it survives the non-linear effects of LLM serving (continuous batching, KV-cache pressure, prefill–decode interference). RQ3 therefore asks: *do the Capacity Units of different request classes add up when the classes are mixed?*

Setup. We ran MoST on 114 mix designs (301 runs) of two to five request classes on the three A40 deployments, in both MIT and MST mode, using two design families: equal request shares and *load-balanced* shares ($p_i \propto C_i$, so that every class consumes the same fraction of the server). 83 designs were repeated two to six times (270 runs) to assess stability. We compare Eq. (1) with the geometric and arithmetic means of the class capacities by embedding all three in the power mean $M_q = (\sum_i p_i C_i^q)^{1/q}$ ($q = -1$ harmonic, $q = 0$ geometric, $q = +1$ arithmetic). Capacities are latent in MoST, since only pass/fail verdicts at the probed rates are observed; the Bayesian analysis therefore models each run's capacity as the threshold of a logistic pass/fail curve, fitted in two stages (pure classes first, mixes second) with design and run random effects. The frequentist analysis uses the final binary-search bracket and takes the mix design as the unit of inference.

The harmonic law holds. The data select the harmonic law (Table 1, Fig. 1): the estimated exponent is $q = -0.96$ (95 % CrI $[-1.02, -0.88]$; profile likelihood -0.91, 95 % CI $[-0.98, -0.84]$), the harmonic prediction is closer than the arithmetic one for 97 % of the runs, and leaving q free does not improve the

out-of-sample predictive density ($+1.7 \pm 3.3$ log units), whereas the geometric (-264.5) and arithmetic (-441.4) laws are clearly worse. A naive arithmetic average of capacities overestimates mix capacity by $-53\,\%$ on average. Regressing log measured capacity on the log harmonic prediction gives a slope of 0.96 (95 % CI $[0.90, 1.02]$, $R^2 = 0.97$). Mix capacity is also bounded by its classes: 294 of 301 runs lie between the capacities of their slowest and fastest class, and none exceeds the arithmetic mean.

A small, model-dependent bias. Mixes deliver slightly less than their Capacity Units promise: $-5.5\,\%$ on average (90 % CrI $[-9.9\,\%, -1.0\,\%]$; frequentist $-3.0\,\%$, 95 % CI $[-6.9\,\%, +1.0\,\%]$), which is practically equivalent to zero within $\pm 10\,\%$ (TOST $p = 0.001$; $P(|\delta| < 10\,\%) = 92\,\%$). Table 2 breaks the bias down by model and test. In SCALE terms, an oversubscription ratio ρ applied on top of an accounting that is $-5.5\,\%$ optimistic commits about $\rho/(1+\delta)$ of the physical capacity; providers should either reserve a margin of a few percent on SK_{max} or account for it when setting ρ.

Same law, different precision in MIT and MST. Both MoST tests follow the same law with the same mean bias ($-2.9\,\%$ vs. $-3.1\,\%$, $p = 0.507$), but individual predictions are about twice as precise for the 2-minute MIT test as for the 30-minute MST test (mean absolute error 10 % vs. 23 %, Levene $p = 0.0005$; Table 3, Fig. 2). Repeated runs are very stable (0.3 % and 1.8 % between repeats of a design, 0.3 % and 1.7 % between repeats of a pure class), so the spread reflects genuine design-specific departures (6 % in MIT, 21 % in MST) rather than measurement noise. A 90 % planning band around Eq. (1) is $\pm 15\,\%$ for MIT and $\pm 39\,\%$ for MST. This matters for SCALE because its user-facing currency Q is derived from the sustainable capacities, precisely where additivity is least precise. Because the departures are tied to the design and not to noise, they do not average out across users who send the same kind of mix.

The heaviest class breaks additivity, depending on the model. Without the heaviest class (600–1000 input and 1000–1500 output tokens), the three models behave alike (ANOVA model effect $p = 0.544$) and 75 % of the MIT designs are predicted within $\pm 10\,\%$. With it, the model matters ($p = 0.0007$; Table 4): Gemma mixes depart from the prediction by $+64\,\%$ in MST and Llama mixes by $-24\,\%$. The share of designs within $\pm 10\,\%$ drops to 42 % (MIT) and 25 % (MST). Two effects are consistent with this pattern. First, the KV-cache footprint per token differs widely (about 448 KB for Gemma-7B with multi-head attention versus 128 KB and 56 KB for the grouped-query attention of Llama-3.1-8B and DeepSeek-R1-Distill-Qwen-7B), which affects heavy requests most. Second, MoST evaluates its MST criteria (response-time trend, backlog, success rate) over all the requests of a run: in a mix, the many short requests can mask the slow ones. Gemma's heavy class alone fails the MST criteria at 8 req/min while completing every request, yet equal-share mixes containing it pass at about 40 req/min. Resources may thus add up while the QoE verdict does not, which argues for per-class QoE criteria when MoST is applied to mixed workloads.

Table 1. Combination laws for mix capacity. Errors are log-ratios of measured to predicted capacity averaged over the 114 mix designs; the last column is the out-of-sample log predictive density relative to the harmonic law (± standard error).

Law	q	Mean error	RMSE	Within ±10 %	Δ log pred. density
Harmonic	-1	-3.0 %	25 %	55 %	0 (reference)
Geometric	0	-38.7 %	79 %	8 %	-264.5 ± 24.8
Arithmetic	$+1$	-52.6 %	156 %	4 %	-441.4 ± 38.1
Free exponent	-0.96 [-1.02, -0.88]	–	–	–	$+1.7 \pm 3.3$

Table 2. Bias of mix capacity relative to the harmonic prediction, by model and MoST test.

			Frequentist	Bayesian	
Model	Test	Designs	Mean [95 % CI]	δ [90 % CrI]	$P(\lvert\delta\rvert < 10\,\%)$
DeepSeek-R1-Distill-Qwen-7B	MIT	23	-7.8 % [-10.7 %, -4.9 %]	-8.8 % [-11.2 %, -6.4 %]	57 %
DeepSeek-R1-Distill-Qwen-7B	MST	15	-3.0 % [-12.5 %, $+7.6$ %]	-4.1 % [-10.2 %, $+3.1$ %]	91 %
Gemma-7B	MIT	23	$+5.1$ % [-1.2 %, $+11.8$ %]	-0.3 % [-3.7 %, $+3.1$ %]	100 %
Gemma-7B	MST	19	$+8.7$ % [-11.1 %, $+32.9$ %]	-4.6 % [-10.0 %, $+2.0$ %]	91 %
Llama-3.1-8B-Instruct	MIT	16	-6.8 % [-9.4 %, -4.1 %]	-7.0 % [-9.6 %, -4.4 %]	91 %
Llama-3.1-8B-Instruct	MST	18	-14.4 % [-21.7 %, -6.3 %]	-7.8 % [-14.2 %, -2.4 %]	64 %
All	both	114	-3.0 % [-6.9 %, $+1.0$ %]	-5.5 % [-9.9 %, -1.0 %]	92 %

1.2 Burst factor (RQ4)

SCALE exposes a single currency based on the sustainable capacity and keeps the instantaneous capacity IK_{max} as a background safety threshold for bursts. We define the *burst factor* of a request class as the ratio of its instantaneous to its sustainable throughput,

$$B_r = \frac{MIT_r}{MST_r}, \tag{2}$$

i.e., how many times its sustainable rate a client may reach during a 2-minute burst. Since $IK_r/SK_r = (IK_{max}/SK_{max})/B_r$, the sustainable and instantaneous Capacity Units rank and price request classes identically if and only if B_r is constant across classes. A constant burst factor is therefore what justifies exposing one currency. RQ4 asks: *is the burst factor constant across request classes and deployments, and can it be used to size the burst allowance of a plan?*

The burst factor is nearly constant. Across the 60 request classes of the three deployments, the median burst factor is 3.50 (interquartile range 3.41–3.94; Table 7, Fig. 4), and the per-model medians are almost identical: 3.57 for DeepSeek (95 % CrI $[3.49, 3.64]$), 3.56 for Gemma ($[3.49, 3.65]$) and 3.55 for Llama ($[3.48, 3.62]$). The model has no detectable effect (ANOVA $p = 0.742$): the posterior ratio between the burst factors of any two models lies within 3.5 % of one (95 % CrI), and the probability that they are equivalent within ±10 % is at least 100 %. Moreover, 77 % of the classes lie within ±15 % of their model's median. The

Table 3. MIT versus MST: the same law with different precision.

	MIT	MST	Test
Mean bias	−2.9 %	−3.1 %	p = 0.507
Mean absolute error	10 %	23 %	p = 0.020
SD of log error	0.12	0.30	Levene p = 0.0005
Within ±10 %, all mixes	61 %	48 %	
Within ±10 %, without heavy class	75 %	58 %	Levene p = 0.007
Within ±10 %, with heavy class	42 %	25 %	
σ_{rep} (repeat pure cell)	0.3 %	1.7 %	
σ_{run} (repeat mix design)	0.3 %	1.8 %	
σ_{mix} (design departure)	6 %	21 %	

Table 4. Mean departure from the harmonic law by model, with and without the heaviest request class.

	Without heavy class		With heavy class	
Model	MIT	MST	MIT	MST
DeepSeek-R1-Distill-Qwen-7B	−4 %	−1 %	−12 %	−8 %
Gemma-7B	−1 %	−10 %	+16 %	+64 %
Llama-3.1-8B-Instruct	−5 %	−9 %	−9 %	−24 %
ANOVA model effect	p = 0.544		all mixes: p = 0.0007	

burst factor is thus a highly portable property of a MoST-characterised deployment, which strengthens the portability claim of RQ1 with a quantity that can be compared across models in absolute terms, unlike K.

Systematic exceptions. Once Gemma's two heaviest-output cells are set aside, the output length has a clear effect ($p = 6 \times 10^{-7}$): classes with short answers (1–100 output tokens) tolerate larger bursts ($B = 4.15$ vs. 3.68, +13 %, $p = 0.0004$). The two Gemma cells with long inputs and long outputs reach B up to 11.6; they coincide with the MST anomaly discussed in Sect. 1.1. For mixes measured in both tests (12 designs), the measured burst factor exceeds the one predicted from the class capacities by +19 % on average, because mixes depart more from additivity in MST than in MIT (Fig. 5).

Implications for SCALE. A single burst factor of about 3.50 supports SCALE's design choice of one user-facing currency: pricing bursts with sustainable Capacity Units mis-states their instantaneous load by less than 15 % for 77 % of the classes. It also fixes the parameters of the Token Bucket that SCALE leaves open (Sect. 1.3). Note that the illustrative capacity profile of the introduction (30,000 requests per 30 minutes and 1,400 requests per minute, i.e., $B \approx 1.4$) is far more conservative than the measured burst factor.

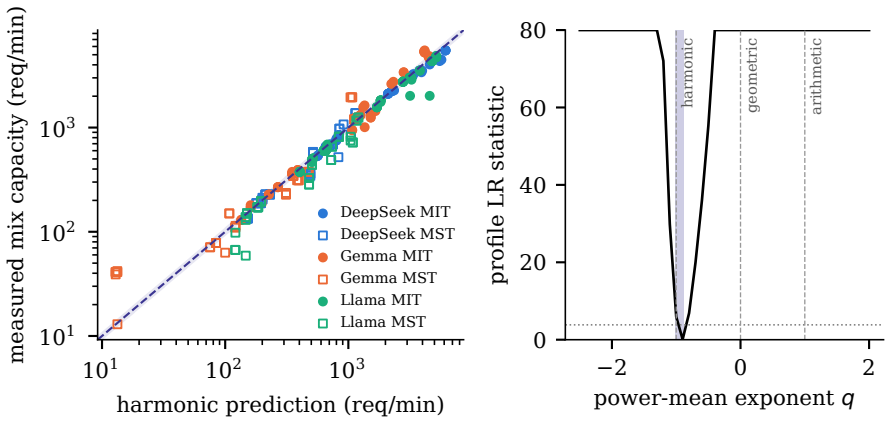


Fig. 1. Left: measured mix capacity versus the harmonic prediction of Eq. (1) (filled: MIT, hollow: MST; band: ±10 %). Right: profile likelihood of the power-mean exponent q, with the Bayesian 95 % credible interval shaded.

1.3 Sizing the token bucket from the burst factor

SCALE refills each user's Consumption Units with a Token Bucket [?], but leaves its two parameters open: the refill rate s follows from the plan's share of SK_{max}, whereas the bucket depth b, which determines how large and how long a burst can be, is not specified. The burst factor closes this gap with three rules expressed in the single sustainable currency.

Rule 1: bucket depth. A client that bursts at B times its refill rate for the MIT window W spends $B\,s\,W$ units and receives $s\,W$, so the bucket must hold

$$b \;=\; (B-1)\,s\,W. \tag{3}$$

With $B = 3.50$ and $W = 2\,\text{min}$, the bucket should hold about 5.0 minutes of refill (4.6 minutes with the conservative $B_{safe} = 3.29$, the 10th percentile over classes, which covers 90 % of them). Because B is the same for every class and model, this single rule applies to all plans; plans restricted to short requests (up to 300 tokens, $B = 4.05$) can use 6.1 minutes.

Rule 2: throttling threshold in the same currency. SCALE keeps IK_{max} as a background safety threshold. Since $IK_r/SK_r = (IK_{max}/SK_{max})/B$, that threshold can be expressed in sustainable units as

$$\Theta \;=\; B \cdot SK_{max} \quad \text{per minute, averaged over the last } W \text{ minutes}, \tag{4}$$

e.g., 8,226, 7,297 and 7,717 units per minute for the DeepSeek, Gemma and Llama deployments (Table 5). No second accounting in IK is needed.

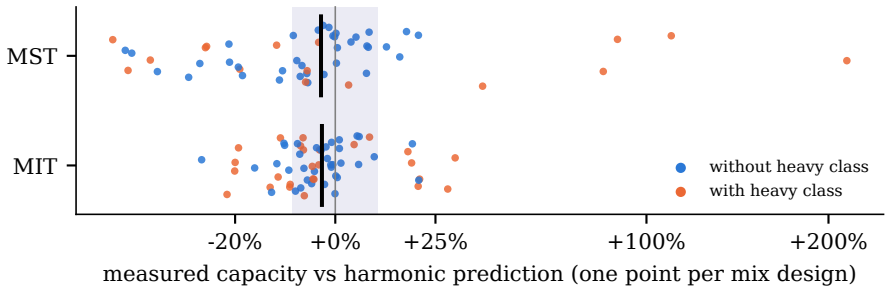


Fig. 2. Departure of every mix design from the harmonic law in MIT and MST (black tick: mean; band: ±10 %). Orange designs contain the heaviest request class.

Rule 3: burst-safe oversubscription. Oversubscription and bursts compound: if a share φ of an allocation oversubscribed by ρ bursts at once while the rest runs at its sustainable rate, the instantaneous load is $\rho\,(\varphi B + 1 - \varphi)\,SK_{max}$, which stays below Θ iff

$$\rho \;\leq\; \frac{B}{1 + \varphi\,(B-1)}. \tag{5}$$

With $\rho = 1.2$, up to 77 % of the allocation can burst simultaneously; if at most half of it bursts at once, ρ can reach 1.56 (Fig. 3). Equation (5) turns the statistical bet of oversubscription into an explicit condition on burst concurrency that a provider can monitor.

Application to the illustrative plans. Table 6 re-derives the plans of Table 3 from the measured burst factor. A Basic user (50 requests per 30 minutes) can be granted bursts of about 14 requests per 2 minutes with a bucket of 10 requests, and a Pro user (175 per 30 minutes) about 47 with a bucket of 36. The burst values assumed in Table 3 correspond to 21 times the sustained rate, far above the measured B; they are safe only if bursts rarely coincide, which Eq. (5) makes explicit.

Contribution. Together, Eqs. (3)–(5) turn one measured, portable constant into the missing parameters of SCALE's admission control: the bucket depth of every plan, the throttling threshold in the user-facing currency, and a burst-aware bound on the oversubscription ratio.

Threat to validity. The constancy of B may partly reflect MoST's measurement windows (2 versus 30 minutes) and thresholds rather than the hardware alone. Repeating MIT with other window lengths on a few classes would tell whether B, and hence Eq. (3), should be reported per window length.

Table 5. Token-bucket parameters derived from the burst factor: typical and conservative (B_{safe}, 10th percentile over classes) burst factor, bucket depth in minutes of refill (Eq. (3)), throttling threshold in sustainable units per minute (Eq. (4)) and the largest share of the allocation that may burst at once for $\rho = 1.2$ (Eq. (5)).

	Burst factor		Depth b/s (min)				
Model	B	B_{safe}	typical	safe	SK_{max}	$\Theta = B \cdot SK_{max}$	φ_{max} ($\rho = 1.2$)
DeepSeek-R1-Distill-Qwen-7B	3.57	3.31	5.1	4.6	2,307	8,226	77 %
Gemma-7B	3.56	3.28	5.1	4.6	2,047	7,297	77 %
Llama-3.1-8B-Instruct	3.55	3.38	5.1	4.8	2,176	7,717	77 %

Table 6. The illustrative plans of Table 3 re-derived from the measured burst factor of requests of up to 300 tokens ($B = 4.05$).

		Quota Sustained s	Derived from B		Table 3
Plan	ρ	(req/30 min)	(req/min)	Burst (req/2 min)	Depth b (req)	burst (req/2 min)
Basic	1.0	50	1.67	14	10	70
Basic	1.2	60	2.00	16	12	70
Pro	1.0	175	5.83	47	36	245
Pro	1.2	210	7.00	57	43	245

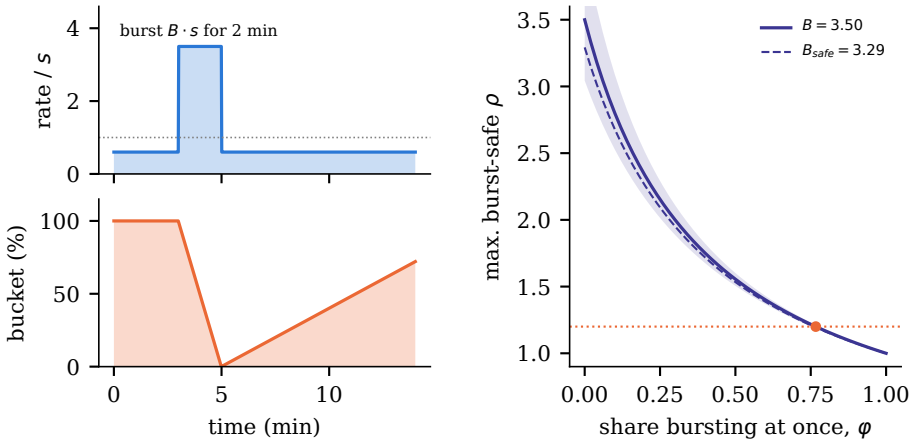


Fig. 3. Left: a client with refill rate s bursts at $B \cdot s$ for W minutes, which empties a bucket of depth $(B-1)\,sW$ (Eq. (3)). Right: largest burst-safe oversubscription ratio as a function of the share of the allocation bursting at once (Eq. (5)); band: $B \pm 15\,\%$; dot: $\rho = 1.2$.

Table 7. Burst factor $B = MIT/MST$ per model: median over the request classes (Bayesian 95 % credible interval), interquartile range, range, and share of classes within ±15 % of the median.

Model	Median B [95 % CrI]	IQR	Range	Within ±15 %
DeepSeek-R1-Distill-Qwen-7B	3.57 [3.49, 3.64]	3.42–4.13	3.22–5.63	70 %
Gemma-7B	3.56 [3.49, 3.65]	3.40–3.90	2.99–11.64	75 %
Llama-3.1-8B-Instruct	3.55 [3.48, 3.62]	3.42–3.76	3.25–5.37	85 %
All	3.50	3.41–3.94	–	77 %

Table 8. Burst factor per request class. Bold: more than 15 % away from the model's median.

		Output tokens				
Model	Input	1-100	100-300	300-600	600-1000	1000-1500
DeepSeek	600-1000	**4.62**	3.60	3.41	3.44	3.49
	300-600	3.93	**4.56**	3.22	3.32	3.27
	100-300	**4.65**	**4.11**	**4.19**	3.44	3.43
	1-100	**5.63**	3.60	3.47	3.51	3.41
Gemma	600-1000	3.45	**4.08**	3.48	3.62	**11.64**
	300-600	3.64	3.65	3.43	3.28	**6.02**
	100-300	3.98	3.87	3.41	3.26	3.40
	1-100	**4.24**	3.37	3.46	3.37	**2.99**
Llama	600-1000	3.91	3.56	**4.13**	3.52	3.29
	300-600	3.58	3.71	3.43	3.41	3.25
	100-300	3.99	**5.37**	3.48	3.39	3.43
	1-100	**4.69**	3.55	3.45	3.50	3.41

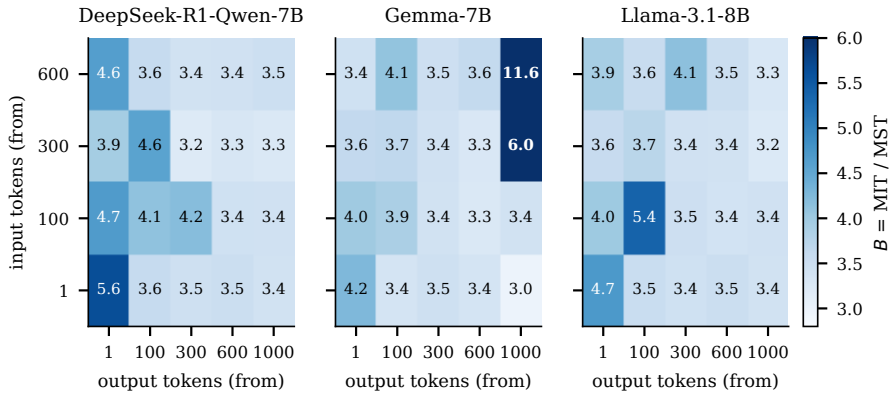


Fig. 4. Burst factor $B = MIT/MST$ of every request class (rows: input tokens; columns: output tokens; labels give the lower bound of each interval).

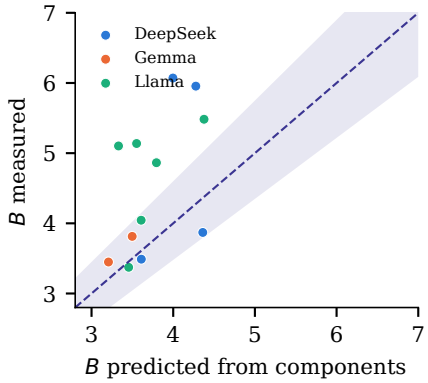


Fig. 5. Burst factor of mixes measured in both tests versus the value predicted from their classes (band: ±15 %).